MILITARY SPECIFICATION

CARTRIDGE, 20MM, HIGH EXPLOSIVE INCENDIARY WITH TRACER, M242

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electric primed cartridge for use in Guns, 20mm, Automatic. (see 6.1)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition, General Specifications for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-644 - Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing, and Marking of Small Arms Ammunition
MIL-STD-651 - Visual Inspection Standards for 20mm Ammunition and Components
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMAMENT COMMAND

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

2.1 (Cont)

D7258928 - Marking on Shipping and Storage Container, M548, for 20mm Cartridges
F7258941 - Method of Packing 100 Cartridges, 20mm, with M103 Type Case. In Bulk, In M548 Container
F7258942 - Method of Packing 200 Cartridges, 20mm, with M103 Type Case, In Bulk, In M548 Container
F7258968 - Method of Packing 100 Cartridges, 20mm, In M12 Type Links, In M548 Container
F7258969 - Method of Packing 100 Cartridges, 20mm, In M14 Type Links, In M548 Container
F7258970 - Method of Packing 100 Cartridges, 20mm, In M17 Type Links, In M548 Container
D7259237 - Marking on Shipping and Storage Container, M548, For Cartridges, 20mm, Linked (Functional)
D7259268 - Cartridge, 20mm, High Explosive Incendiary With Tracer, M242
IEL 7259268 - Index of Inspection Equipment Lists For: Cartridge, 20mm, High Explosive Incendiary With Tracer, M242

PUBLICATIONS

U.S. ARMY MATERIEL COMMAND

AMCR 715-505 - Ammunition Ballistics Acceptance Test Methods, Volume 8, Test Procedures for 20mm Cartridges

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-The cartridge shall comply with Drawing (Dwg.) D7259268, referenced specifications, and other requirements specified herein.

3.2 Manufacturing Process.-All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without his prior approval. (see 6.3)

3.3 First Article Sample -Unless otherwise directed by the contracting officer, a first article sample shall be required.

3.4 Projectile Extraction.-The force necessary to extract the projectile from the cartridge case shall be as specified by Dwg. D7259268.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

3.5 Electrical Resistance, Primed Case.-The electrical resistance of the primed case shall not be less than 500 ohms or more than 1,200,000 ohms.

3.6 Projectile Torque.-The projectile shall withstand a minimum torque as specified by Dwg. D7259268.

3.7 Velocity.-The average velocity of the sample cartridges conditioned at 68 degrees Fahrenheit (^{o}F) to 72°F shall be 3380 $\pm$ 50 feet per second (fps) at 78 feet from the gun muzzle and the sample standard deviation shall not exceed 40 fps.

3.8 Pressure.-The average chamber pressure ($\bar{X}$) of the sample cartridges, conditioned at 68°F to 72°F, plus 0.84 times the average ($\bar{R}$) of the chamber pressure shall not exceed 60,500 pounds per square inch (psi).

3.9 Waterproofness.-The average velocity (wet) of projectiles of the sample cartridges conditioned at 68°F to 72°F shall not vary from the average velocity (dry) by more than 100 feet per second.

3.10 Accuracy.-The average mean radius of all targets of the sample cartridge projectiles at ambient temperature shall not exceed 15 inches at a range of 600 yards.

3.11 Action Time.-The action time of the cartridge shall not exceed 4.0 milliseconds at either ambient or -65°F temperature.

3.12 Projectile Trace.-The projectile shall exhibit a visible trace, when viewed against a dark background, upon ignition of the trace mixture and continue to trace for a minimum of 1.9 seconds after cartridge ignition.

3.13 Projectile Function.-The projectile shall function with high order detonation upon target impact.

3.14 Function and Casualty.-The cartridge, at ambient temperature, shall function without casualty.

3.15 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications, and the following:

3.15.1 Process Defects.-The cartridge shall be free of defects such as folds, wrinkles, buckles, deep draw scratches, cuts, dents, perforations, and other forms of substandard workmanship.

3.15.2 Cleanliness.-The cartridge and its component shall be free of all foreign matter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance Terms and Definitions.-Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 Classification of Inspections.-The inspection requirements specified herein are classified as follows:

1. First Article Inspection (see 4.3).
2. Quality Conformance Inspection (see 4.4).

4.3 First Article Inspection.

4.3.1 First Article Sample.-The sample shall consist of 680 cartridges and shall be delivered for inspection in accordance with the first article approval provisions of the contract and MIL-A-2550.

4.3.1.1 Inspection.-The first article sample will be subjected to the Quality Conformance Inspection provisions as stated herein, except that approval will be based upon the examination of the complete sample for visual, dimensional characteristics of the drawings and related documents, and the initial test sample criteria in accordance with Table I.

4.4 Quality Conformance Inspection.

4.4.1 Submission of Product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.4.1.1 Lot.-A lot shall consist of:

a. Cartridges loaded by one manufacturer, in one unchanged process, in accordance with the same drawings and drawing revisions, and the same specification and specification revision.

b. Like parts and assemblies (cases, primers, projectiles) having one manufacturer's symbol and one interfix number.

c. Propellant from one lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

4.4.1.2 Lot Identification -Each cartridge and each packed ammunition lot shall be identified in accordance with applicable drawings and MIL-STD-1168. Each packed lot shall further be identified by a National stock number assigned by the procuring activity.

4.4.2 Examination -One hundred percent examination shall be performed by the contractor for all critical defects. Examination for major and minor defects shall be performed as specified herein. Each part and each assembly found to be defective shall be rejected.

4.4.2.1 Primed Cartridge Cases.-Each primed cartridge case shall be tested by the contractor for electrical resistance subsequent to primer crimping and waterproofing, using the method of 4.5.2.1. Primed cases which fail to meet the requirement of 3.5 shall be removed from the lot.

4.4.2.2 Sampling and Acceptance Criteria.-The sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235. To determine product acceptability, major or minor defects as listed herein may be considered collectively on a class basis or they may be considered individually. However, where major defects are considered collectively on a class basis, the acceptance number for any individual defect of the class shall be limited by the assigned Acceptable Quality Level (AQL) for individual defect associated with the class. (see 4.4.2.2.1.c) In addition, where MIL-STD-1235 is used, sampling plans applicable to a class shall not be used for major defects. (see 6.4)

4.4.2.2.1 AQL's.-The following AQL's are assigned to major and minor defects:

a. Individual:

Major	0.25%
Minor	0.40%

b. Class basis:

Major	1.50%
Minor	2.50%

c. Individual defects associated with the class basis:

Major	0.40%

4.4.3 Classification of Defects 1/.-The classification of defects shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

4.4.3.1 Charged Body Projectile, Prior to Assembly of Seal (see Dwg. D7259270, a detail of D7259268).

Categories and Defects	Method of Inspection
Critical:	
1. Tracer charge, missing	Visual
Major: None defined.	
Minor: None defined.	

4.4.3.2 Charged Body Projectile (see Dwg.D7259270, a detail of D7259268.)

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Metering disc missing	Visual
102. Loose metering disc	Manual 2/
103. Seal missing or perforated	Visual
104. Foreign matter	Visual
Minor: None defined.	

4.4.3.3 Cartridge (see Dwg. D7259268).

Categories and Defects	Method of Inspection
Critical:	
1. Split or perforated case	Visual
2. Crack or split in projectile	Visual
3. Crimp missing (case-projectile)	Visual
Major:	
101. Loose primer	Manual
102. Depth of primer seating	Gage
103. Over-all length, max	Gage
104. Improper crimp (case-projectile)	Visual
105. Profile and alignment, max	Gage 3/
106. Corrosion	Visual
107. Head configuration	Visual
108. Metal slivers on head face	Visual
109. Length, case head to centerline crimp	Gage
110. Crack, split, or dent in fuze nose cap	Visual
111. Fuze loose	Gage 4/

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

7/ Pressure.-If the average pressure ($\bar{X}$) plus 0.84 times the average range ($\bar{R}$) of the sample cartridges exceeds 60,500 psi, a second sample of 40 cartridges shall be tested. The lot shall be rejected if the average pressure ($\bar{X}$) plus 0.91 times the average range ($\bar{R}$) of the retest sample fails to comply with the applicable requirement. (see 6.6.2)

8/ Waterproofness.-If the average velocity (wet) of the sample fails to comply with the applicable requirement, a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the average velocity (wet) differs by more than 100 fps from the average velocity (dry) of 40 additional sample cartridges fired to establish a new base for comparison. The lot shall not be penalized for failure of the last named sample cartridges to comply with the requirement of 3.7.

9/ Accuracy.-If the average mean radius of the sample cartridges exceeds the applicable requirement, a retest sample of 100 cartridges shall be tested. The lot shall be rejected if the average mean radius of ten accuracy targets exceeds 15 inches at the same range.

10/ Action Time.-If any cartridge of the test sample exhibits an action time exceeding 4 milliseconds (0.004 seconds) or the average action time ($\bar{X}$) of the sample cartridges plus 4 sample standard deviations exceeds 4 milliseconds, the lot shall be rejected. These criteria shall apply independently to each temperature phase of the test.

11/ Projectile Trace.-Failure of 7 or more sample cartridges to comply with applicable requirements shall cause rejection of the lot. If more than 3 but less than 7 fail to comply, a second sample of 50 cartridges shall be tested. The lot shall be rejected if 9 or more cartridges of the combined sample fail to comply with the minimum requirement.

12/ Projectile Function.-Failure of 5 or more sample cartridges to comply with the applicable requirement shall cause rejection of the lot. If more than 2 but less than 5 fail to comply, a second sample of 50 cartridges shall be tested. The lot shall be rejected if 7 or more cartridges of the combined samples fail to comply with the applicable requirement.

13/ Function and Casualty.-The lot shall be rejected if any malfunction or firing casualty of Table II occurs in number(s) equal to or greater than the applicable "Rej" number. Except as otherwise provided, if malfunctions or casualties occur in excess of the applicable "Acc" number, but less than the applicable "Rej" number, a second sample of 800 cartridges shall be divided and fired in equal numbers in the M39 and M61 guns. The lot shall be rejected if in the combined samples, malfunctions or casualties occur in numbers equal to or greater than the applicable "Rej" number.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

14/ Additional cartridges are not required for these tests as cartridges from other test samples may be used as follows:

a. Primed cases and propellant from the projectile extraction test are used as samples for the electrical resistance verification and propellant contamination tests.

b. Velocity and pressure tests are fired concurrently.

4.4.4.2 Firing Defects.-Firing defects and associated criteria shall be as specified in Table II. For defect definitions, see AMCR 715-505, Volume 8.

Table II

Firing Casualties	Acc	Rej
Cartridge:		
Fire "out of battery", M61 gun 1/	0	1
Misfire 2/	-	-
Failure to chamber	0	1
Failure to extract	0	1
Projectile remaining in bore	0	1
Primer:		
Missing primer or button missing from primer	0	2
Primer leak	15	40
Case:		
Longitudinal split 3/		
I or S	10	31
J	4	11
K	0	2
L or M	0	1
Circumferential rupture, partial 3/		
S, J or K	4	11
L	0	1
Circumferential rupture, complete 3/	0	1
Detached metal 4/	0	1
Projectile:		
Rotating band separation 5/		
Complete or partial separation having band seat serration imprint	0	3
Separation having length greater than 0.5 inch, and without band seat serration imprint	1	6
Metal parts separation, except rotating band separation 6/	0	1
Premature functioning	0	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

Minor:

201. Marking incorrect, incomplete, illegible or missing -- Visual
202. Projectile protective coating damaged or incomplete -- Visual
203. Waterproofing missing around primer ---------- Visual
204. Scratch, dent, buckle, bulge, wrinkle or fold in case - Visual
205. Rotating band damaged ---------- Visual
206. Primer crimp missing ---------- Visual
207. Foreign matter, except corrosion ---------- Visual
208. Incorrect type cartridge ---------- Visual
209. Gap between fuze and projectile body ---------- Gage
210. Lacquer on primer button ---------- Visual (see 6.5)

1/ MIL-STD-651 shall apply in defining and evaluating cartridge visual defects.

2/ The projectile shall be classed defective if the metering disc can be moved laterally in its seat.

3/ A dead weight load of 60 pounds maximum may be used to insert the cartridge into the gage.

4/ When the fuze turns with respect to the projectile body upon application of the specified torque in a counterclockwise direction, the cartridge shall be classed defective.

4.4.4 Tests.-The tests listed in Table I shall be performed on each cartridge lot in accordance with the test methods prescribed in 4.5. Sample size and acceptance criteria for each test shall be as specified. Only cartridges which have met the visual and dimensional requirements and which have been selected in such a manner that the sample is representative of the entire lot shall be used in the tests. The selected cartridges shall be mixed thoroughly before being divided into samples for the various tests. (see 6.7)

Table I

Test 1/		Sample Size		Rqmt Para
Projectile extraction	2/	20		3.4
Electrical resistance verification	3/	20	14/	3.5
Propellant contamination	4/	20	14/	3.15.2
Projectile torque	5/	20		3.6
Velocity	6/	20		3.7
Pressure	7/	20	14/	3.8
Waterproofness	8/	20		3.9
Accuracy	9/	50		3.10
Action time	10/			3.11
Ambient temperature		25		
-65°F temperature		25		
Projectile trace	11/	50		3.12
Projectile function	12/	50		3.13
Function and casualty	13/	400		3.14

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

1/ The lot shall be rejected if in any firing test, one or more of the following malfunctions or casualties of Table II occur:

a. Misfire associated with e. or f. of Note 2/ of Table II.
b. Projectile remaining in bore.
c. Metal parts separation, except rotating band separation (see Note 6/ of Table II).
d. Premature functioning.

Except as otherwise specified and in the function and casualty test where the occurrence of a firing defect listed in Table II prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired.

2/ Projectile Extraction.

a. Failure of two or more sample cartridges to conform with the minimum requirement shall cause rejection of the lot. If one cartridge of the sample fails to conform with the minimum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined samples fail to conform with the minimum requirement.

b. Failure of four or more cartridges to conform with the maximum requirement shall cause rejection of the lot. If more than one but less than four cartridges fail to conform with the maximum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if five or more cartridges of the combined samples fail to conform with the maximum requirement.

3/ Electrical Resistance Verification.-The lot will be rejected if any primer of the sample cartridges fails to comply with either or both of the primer resistance requirements.

4/ Propellant Contamination.-The presence of foreign matter in the propellant of any of the test cartridges will cause rejection of the lot. (see 3.15.2 and 6.5)

5/ Projectile Torque.-Failure of two or more of the sample cartridges to comply with the specified torque requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined sample fails to comply with the applicable requirement.

6/ Velocity.-If the sample fails to comply with either or both requirements, a sample of 40 cartridges shall be tested for the failing parameter(s) (average, sample standard deviation). The lot shall be rejected if the retest sample fails to comply with the applicable requirements. (see 6.6.1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

1/ A cartridge shall be classed defective if it fires, due to delayed ignition, after the bolt has been unlocked and extraction has been initiated or completed.

2/ Misfiring test cartridges, excepting those from the waterproofness test (Table I), shall be retested in accordance with the method of 4.5.11. If a cartridge fails to function in that test, examinations shall be conducted for the following conditions:

a. Excessive primer seating depth.
b. Metal slivers across primer insulator.
c. Electrical continuity.
d. Missing or obstructed case vent.
e. Missing or insufficient primer composition or propellant.
f. Partially burned or contaminated primer composition or propellant.

The lot shall not be penalized if condition a. is observed. However, this condition shall be reported for information. The "Acc" number shall be zero (0) and the "Rej" number 3, for misfires associated with conditions b, c, or d. The second sample for evaluation of misfires associated with conditions, b, c, or d shall consist of the function and casualty second sample (see Table I, Note 13/). The lot shall be rejected if either condition e or f is observed. Misfire(s) occurring in the waterproofness test shall be handled in accordance with AMCR 715-505, Volume 8.

3/ See Dwg. C7258801, a detail of IEL 7259268, for classifying splits and ruptures in fired cartridge cases. If a longitudinal split or circumferential rupture (partial) extends into two or more defined areas, only the most severe defect criterion of Table II for the areas involved shall apply. If a rupture results in separation of the cartridge case into two or more portions, the defect shall be classed as a complete circumferential rupture.

4/ Metal sheared or missing from the fired cartridge case exterior, such as rim or neck shears, shall be classed as a defect. The lot shall not be penalized for shavings of metal from the interior wall of the case neck in the crimped area.

5/ For classification as a defect, there must be evidence either by recovery of the band, or portion thereof, or by hole(s) in the fragmentation screen(s). The lot shall not be penalized for normal band fringing, or the separation of band fragments having no evidence of band seat serration imprint and having length of 0.5 inch or less.

6/ Separation or breakup of projectile part(s), except rotating band separation, as evidenced by recovery of the part(s) or fragment(s), or hole(s) in the fragmentation screen(s), shall be classed as a defect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

4.4.4.3 Unlisted Firing Defects.-The lot shall be suspended and referred to the contracting officer for disposition if a malfunction or casualty not covered by this specification occurring in any firing test indicates that the product is unsuited for the purpose intended.

4.4.5 Packaging, Packing and Marking Inspection.-Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644, as applicable to the drawing.

4.4.6 Inspection Equipment.-Index of Inspection Equipment List (IEL) 7259268 identifies the applicable inspection equipment lists indicating the equipment required to perform examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL (reference is made to Inspection Equipment List for definition of coding, Dwg. B11075228, a detail of IEL 7259268). The provisions of MIL-A-2550 shall apply.

4.5 Test Methods and Procedures.

4.5.1 Projectile Extraction.-The method of test shall be as specified in AMCR 715-505, Volume 8.

4.5.2 Electrical Resistance.

4.5.2.1 Primed Cartridge Cases.-The equipment and circuitry specified in IEL 7259268 shall be used to measure the resistance of the primer in each primed cartridge case.

4.5.2.2 Verification.-A Simpson Volt-Ohm-Ammeter, Model 260, or approved equal, shall be used to measure the resistance of the primer in each primed cartridge case from the projectile extraction test sample. Only the RX 100 and RX 10,000 scales of the meter will be used in order to limit the applied voltage and current, respectively, to 7.5 volts and 1.4 milli-amperes, maximum.

4.5.3 Propellant Contamination.-The propellant from each of the cartridges tested for projectile extraction shall be examined visually for contamination as specified in AMCR 715-505, Volume 8.

4.5.4 Projectile Torque.-Equipment shall be as specified in IEL 7259268. Each cartridge of the test sample shall be marked with a light scratch extending axially across the rotating band onto the case neck. Torque as specified in 3.6 shall be applied slowly to the projectile. Movement of the projectile with respect to the cartridge case, detectable by misalignment in the scratch mark, shall be recorded.

4.5.5 Velocity, Pressure and Waterproofness.-The methods of test shall be as specified in AMCR 715-505, Volume 8. Statistics necessary for testing of results against acceptance criteria shall be computed (see 6.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

4.5.6 Accuracy.-The method of test and measurement of targets shall be as specified in AMCR 715-505, Volume 8. The specified range may be shortened in decrements of 100 yards to a minimum range of 200 yards. For ranges less than 600 yards the average mean radius shall be reduced in direct proportion to the reduction in range.

4.5.7 Action Time.-The method of test shall be as specified in AMCR 715-505, Volume 8. Statistics necessary for testing of results against the acceptance criterion shall be computed. (see 6.6)

4.5.8 Projectile Trace.-The method of test shall be as specified in AMCR 715-505, Volume 8, except that observation shall be from near the gun position and projectile trace time will begin at cartridge ignition and continue until the end of the tracer mixture burn.

4.5.9 Projectile Function.-The method of test shall be as specified in AMCR 715-505, Volume 8. The aluminum target (2024T3, T4 or T361) of .040 inch nominal thickness shall be located 200 ± 25 yards from the gun muzzle normal to the line of fire. Observation of projectile functioning shall be made from the camera position.

4.5.10 Function and Casualty.-The method of test shall be as specified in AMCR 715-505, Volume 8. The test sample shall be fired in bursts of 50 cartridges and shall be divided and fired in equal numbers in the M39 and M61 weapons. The weapon barrel(s) shall be at ambient temperature at beginning of test and cooled to ambient after each 100 rounds. The cyclic rate of each burst shall not be less than 1400 shots per minute (SPM) in the M39 weapon and 5500 SPM in the M61 weapon. The cyclic rate for each burst shall be recorded. There shall be observation for a distance of 400 feet from the gun muzzle for premature bursts.

4.5.11 Misfires.-Misfiring cartridges shall be retested in rifle as specified in AMCR 715-505, Volume 8, for the accuracy test. Results of examinations required by Table II shall be recorded.

4.5.12 Defective Weapon.-If a test weapon fails to comply with the minimum cyclic rate during a normal burst operation or is found to be defective during or at the end of burst the test shall be suspended. The weapon shall be corrected or replaced and the test repeated in whole or in part, as indicated. Any firing defect that occurred shall be valid for lot penalty in accordance with Table II unless the defect is found to have been caused by the defective weapon.

5. PREPARATION FOR DELIVERY

5.1 Packing, Level A.-The type of ammunition pack shall be in accordance with instructions from the procuring activity. Linked ammunition shall be packed in accordance with Dwg. F7258968, F7258969, or F7258970. Bulk ammunition shall be packed in accordance with Dwg. F7258941 or F7258942.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

5.2 Marking.-Marking on the packed container shall be in accordance with Dwg. D7258928 or D7259237, as applicable.

6. NOTES

6.1 Intended Use.-These cartridges are intended for use in various automatic weapon systems which are chambered to utilize cartridges with the M103 type case configuration.

6.2 Ordering Data -Invitation for bids and contracts or orders will specify the following:

6.2.1 Title, number and date of this specification.

6.2.2 Place of inspection, if not at place of manufacture.

6.2.3 First article sample requirements. (see 3.3 and 4.3)

6.2.4 Detailed packing and marking instructions. (see 5.1 and 5.2)

6.2.5 Provision for the supply, maintenance and disposition of Government Furnished test equipment for acceptance inspection purposes.

6.2.6 Provision for the submission and approval of the manufacturing process. (see 3.2)

6.3 Process Deviation.-A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which may alter the metallurgical or physical properties of the item.

6.4 AQL's.-The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

6.5 Contamination of Explosive Components.-Particular care should be exercised to avoid contamination of primers and propellant by oil, grease, or other foreign matter since such contamination can cause misfire(s), increase action time and lot rejection.

6.6 Computations.

6.6.1 Standard Deviation.-Where computation of a sample standard deviation is specified for determination of lot acceptance, the method of computation will be $S = \sqrt{\frac{(Xi - X)^2}{n}}$ or equivalent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.

6.6.2 Range Plan.-Where a range plan is specified, the individual values will be recorded in subgroups of 5 in the order in which they are observed. The ranges for the subgroups will be added and the sum of the ranges will be divided by the number of subgroups to determine the average range ($\overline{R}$).

6.7 Combining Tests -With approval of the procuring activity, tests may be performed concurrently on the same sample cartridges provided that the test results are not affected by this procedure. The test facility should make maximum use of this procedure in order to minimize testing costs.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1305-A760

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:00Z
Check the source to verify that this is the current version before use.